Carry or Dispersion?

A live post-mortem of a beta-hedged funding book on crypto perpetuals

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Working paper, November 2026. Revised 7 November 2026.

Abstract

A cross-sectional funding-carry book on 23 crypto perpetual futures goes long the eleven coins with the lowest funding rate, short the eleven with the highest, and hedges its net beta to BTC. In backtests it earns a Sharpe ratio of about 1.32 (Binance, February 2021 to May 2025) and 1.24 (Bybit, August 2022 to May 2025). Run as a live shadow ledger on Bybit for 153 days, it lost 15.3%. That ledger only advanced while its host machine was on, so it rebalanced irregularly; replaying the same period on the intended schedule (every seven days) gives a loss of 4.8% instead. Rebalance timing alone moved a five-month result by 10.5 points, because it changed which coins were held. I decompose every period's profit and loss into a funding (carry) leg and a price leg. In both versions the carry was positive and steady but modest (3.5% a year on the clean schedule, against 7.1% in the Bybit backtest), and the result was decided by the price leg, which is not a hedging failure in the usual sense: its R^2 against BTC is 1% in both. The price leg is a cross-sectional dispersion bet. It carries more than 99.9% of the book's variance in backtests, earns most of its historical return in a single 90-day window, and is not removed by refinements such as inverse-volatility weighting or a second (ETH) hedge factor, whose effect on the Sharpe ratio flips sign between venues. On the clean schedule the live result is unremarkable relative to the backtest ($z = -0.84$). The constructive finding is that nearly all of the carry comes from the short leg, which points to a per-coin delta-neutral construction as the way to isolate it.

1 Introduction

Funding on perpetual futures is one of the few cash flows in crypto markets that is paid on a fixed schedule regardless of direction. A position that is short when funding is positive is paid every eight hours. A natural way to collect it systematically is a cross-sectional book: short the coins whose funding is highest, long the coins whose funding is lowest, and receive the spread. This is the crypto analogue of the carry trades documented across asset classes by Koijen et al. [3].

The difficulty is that high funding is not random. It marks coins that leveraged traders are crowding into, which are typically the most volatile coins with the strongest recent momentum. Shorting them imports a large price exposure. The usual answer is a beta hedge to BTC. The book studied here does exactly that, and its backtest looked attractive: a Sharpe ratio above one on two venues, with a carry leg that on its own is almost riskless.

The recorded live result was a loss of 15.3% in 153 days; on the schedule the strategy actually specifies, the same period gives -4.8%. This paper asks why the book lost money, and why the rebalance schedule mattered so much. The answer to both is not that the carry failed, nor that the hedge was mis-specified, but that the book was never mainly a carry strategy. Its risk, and most of its historical return, sit in a residual price leg that is orthogonal to BTC and ETH and driven by the dispersion of individual coins around the market.

The contribution is a careful decomposition rather than a new signal. Section 2 describes the strategy and data. Section 3 reports the live ledger and shows that its losses are unrelated to BTC. Section 4 shows that the same structure is present in backtests, and that the price leg's historical return is concentrated in a few episodes. Section 5 tests whether better weights or a second hedge factor fix the problem. Section 6 discusses what the evidence does and does not support, and what to build next.

2 Strategy and data

2.1 Funding mechanics

A perpetual future has no expiry. To keep its price close to spot, longs and shorts exchange a funding payment at fixed times (every eight hours on both venues used here). With position weight w_i (fraction of equity, positive for long) and funding rate $f_{i,t}$, the funding P&L over period t is $-w_i f_{i,t}$: when funding is positive, longs pay and shorts receive.

2.2 Book construction

The universe is 23 liquid USDT-margined perpetuals.[1] At each rebalance, every seven days, coins are ranked on their latest funding rate. With L_t the $N = 11$ lowest and S_t the $N = 11$ highest,

$$w_{i,t} = \begin{cases} +1/N & i \in L_t, \\ -1/N & i \in S_t, \\ 0 & \text{otherwise}, \end{cases} \tag{1}$$

so each leg has a gross exposure of one and the book is dollar-neutral before hedging. Its funding income is the cross-sectional spread $\frac{1}{N}\left(\sum_{i\in S_t} f_{i,t} - \sum_{i\in L_t} f_{i,t}\right)$.

The hedge is written generally. Let F be a set of hedge instruments with return vector $\mathbf{F}_t$ and trailing covariance matrix Σ_F, estimated over 90 eight-hour periods (30 days). The book's exposure to factor k is $e_k = \sum_i w_i \operatorname{Cov}(r_i, F_k)$, and the hedge that makes the book jointly neutral to all factors is

$$\mathbf{h} = -\Sigma_F^{-1}\,\mathbf{e}. \tag{2}$$

With $F = \{\text{BTC}\}$ this reduces to the familiar single-beta hedge $h = -\sum_i w_i \beta_i$, which is the live configuration. Using the inverse of Σ_F rather than separate betas matters when factors are correlated, as BTC and ETH are: two separate hedges would remove their common component twice.

2.3 P&L decomposition

Every period's return is split into

$$\pi_t = \underbrace{\textstyle\sum_i -w_i f_{i,t} - \sum_k h_k f_{k,t}}_{\text{carry}} + \underbrace{\textstyle\sum_i w_i r_{i,t} + \sum_k h_k F_{k,t}}_{\text{price}} - \text{cost}_t, \tag{3}$$

with costs of 8 basis points on the turnover of both legs and hedge. Ranks, betas and volatilities at t use information up to $t-1$; P&L is realised over $t \to t+1$.

[1] BTC, ETH, BNB, XRP, ADA, SOL, DOGE, LINK, DOT, LTC, BCH, FIL, NEAR, AVAX, XLM, AAVE, UNI, TRX, ATOM, ETC, SAND, AXS, COMP.



Figure 1: Live shadow ledger on Bybit, 6 June 2026 to 6 November 2026. Top: equity of the unlevered book. Bottom: cumulative carry and price contributions. Carry rises steadily; the price leg determines the outcome.

2.4 Data

Backtests use two venues. *Binance*: hourly closes and funding for all 23 coins, February 2021 to May 2025 (4,731 eight-hour periods, 4.3 years). *Bybit*, the venue of the live test: August 2022 to May 2025 (3,091 periods, 2.8 years). Every variant starts trading only after a 35-day warm-up, so that all variants are evaluated on exactly the same periods.[2]

The live test is a *shadow ledger*: no orders are sent. A scheduled job pulls live Bybit prices and the funding events that actually printed, marks the book to market, accrues funding, and rebalances every seven days. It ran from 6 June 2026 to 6 November 2026 (130 steps). Because the data did not exist when the strategy was fixed, this is a genuine out-of-sample test of both the price behaviour and the assumption that funding accrues as modelled. One implementation detail turned out to matter: the job only ran while its host machine was on, so steps were irregular (from minutes to 23 days apart) and rebalances happened whenever the job next ran after seven days had passed. Section 3.1 quantifies the effect by replaying the same period on the intended schedule.

3 The live ledger

Table 1 and Figure 1 summarise the live result. Equity fell from \$100,000 to \$84,700. The carry leg contributed +2.50% and the price leg −17.31%.

The carry was positive, but modest. Recorded funding income annualised to 6.0%, close to the 7.1% the same configuration earned in the Bybit backtest. That comparison flatters the carry: on the clean schedule of Section 3.1 it annualises to 3.5%, about half the backtest rate.

[2] An earlier version of this comparison let the inverse-volatility variants sit flat during the first week of January 2021, an extreme week for altcoins, while the baseline traded it. That flattered them by about 0.13 Sharpe. The warm-up removes the asymmetry.

Days live / steps	153 / 130
Cumulative return	−15.3%
of which carry	+2.50%
of which price	−17.31%
Annualised log return	−39.6%
Annualised volatility	29%
Sharpe ratio (time-normalised)	−1.35
Carry, annualised (Bybit backtest: 7.1%)	6.0%
Std. of price P&L per step	1.64%
Std. of carry P&L per step	0.052%
Worst / best price step	−9.53% / 3.98%

Table 1: Live summary. Steps are irregular, from minutes to 23 days apart, so annualised figures use time-normalised estimators: return $\sum_j \ell_j/T$ and volatility $\sqrt{\sum_j \ell_j^2/T}$, with ℓ_j the log return of step j and T the elapsed time in years.

The two ledgers earn nearly the same carry in every month except October (+1.43% recorded against +0.46% clean), when the recorded book happened to hold coins with unusually high funding. The cash flow is real and accrues as modelled, but it is smaller than the backtest suggested.[3]

The price leg dominated. Per step, the price P&L was 31 times more volatile than the carry P&L. There were 4 steps with a price move above 4%, the worst at −9.53% and −8.14%. A book that is genuinely market-neutral should not produce such steps.

It was not BTC. The obvious suspect is a hedge that failed to neutralise the market. Regressing each step's price P&L on the BTC log return over the same interval (Figure 2) gives a slope of 0.06 and an R^2 of 1%. Adding ETH raises the R^2 to 2%. The worst step, ending 21 September, coincided with a BTC move of +10.1%: the beta hedge gained, but the shorted high-funding coins rallied much harder. The second worst, ending 9 September, came with BTC at only −1.8%. The hedge did its job. What remained is the dispersion of individual coins around the market, which a hedge in BTC, or in BTC and ETH, cannot touch by construction.

3.1 How much did the schedule matter?

The live ledger rebalanced 16 times at irregular moments; the strategy specifies a rebalance every seven days. Because the ledger is a shadow book, the intended schedule can be reconstructed exactly from Bybit's public history: every eight-hour funding time is marked to the close of the hour ending then, funding printed in the interval is accrued, and the book is rebalanced every seven days using only data available at that moment.[4] Since 7 November 2026 the live job itself works this way, so missed runs are caught up and no longer change the result.

Table 2 and Figure 3 show that the schedule moved the five-month result by 10.5 points. Nothing about the strategy's rules changed; only the days on which it re-ranked the coins. Rebalancing on a different day selects different coins: at the end of the window only 5 of the 10 long positions (excluding BTC) are the same in the two books. In a book whose outcome is decided by which coins are held during a few large dispersion moves, that is enough to turn −4.8% into −15.3%. The sensitivity is itself evidence for the paper's thesis: a carry strategy

[3] The live script quotes a higher expectation (9.2%) taken from an earlier research notebook with a different sample.

[4] The replay uses the same weighting code as the live job and was tested for two properties: running it in one pass or in several passes with gaps gives identical ledgers, and the weights at a rebalance are identical whether or not data after that moment is supplied.

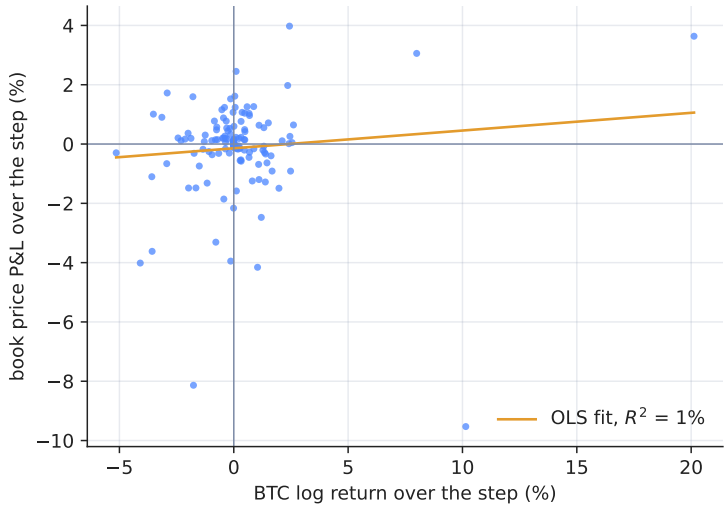


Figure 2: Live price P&L per step against the BTC return over the same interval. The beta hedge works ($\beta \approx 0.06$); the large losses are unrelated to the market.

	Recorded ledger	Clean schedule
Rebalances	16	21
Cumulative return	−15.3%	−4.8%
of which carry	+2.50%	+1.46%
of which price	−17.31%	−5.13%
Carry, annualised	6.0%	3.5%
Sharpe ratio	−1.35	−0.49
Price P&L R^2 against BTC	1%	1%

Table 2: Recorded live ledger against the same period replayed on the intended schedule (clean replay to 6 November, 08:00 UTC, the last eight-hour mark before the end of the recorded window).

would barely notice the rebalance day; a dispersion bet does. The diagnosis of the previous paragraphs is unaffected: on the clean schedule too, the price leg dominates and is unrelated to BTC.

Is the loss significant? Under i.i.d. returns the standard error of an annualised Sharpe ratio estimated over T years is approximately $\sqrt{(1 + \mathrm{SR}^2/2)/T}$ [4]. With $T = 153/365$ and the Bybit backtest Sharpe of 1.24 as the null, the standard error is 2.05. The recorded Sharpe of −1.35 gives $z = -1.26$; the clean schedule's −0.49 gives $z = -0.84$. Neither rejects the backtest. The empirical distribution makes the role of the schedule sharper. Of all 153-day windows in the backtests, 29% (Binance) and 23% (Bybit) lost money. A loss of 15.3% or more, as recorded, occurred in only 3% of Binance windows and in none of the Bybit windows. A result of −4.8% or worse, as on the clean schedule, occurred in 19% and 14% of windows. The recorded loss lies beyond anything the backtest produced; the strategy as specified had an ordinary bad five months. Most of what looked like a tail event was the execution schedule.

4 The same structure in backtests

Figure 4 shows that the live pattern is not new. In both backtests the carry leg is close to a straight line: on its own it has a Sharpe ratio of 22.9 on Binance and 11.2 on Bybit, with

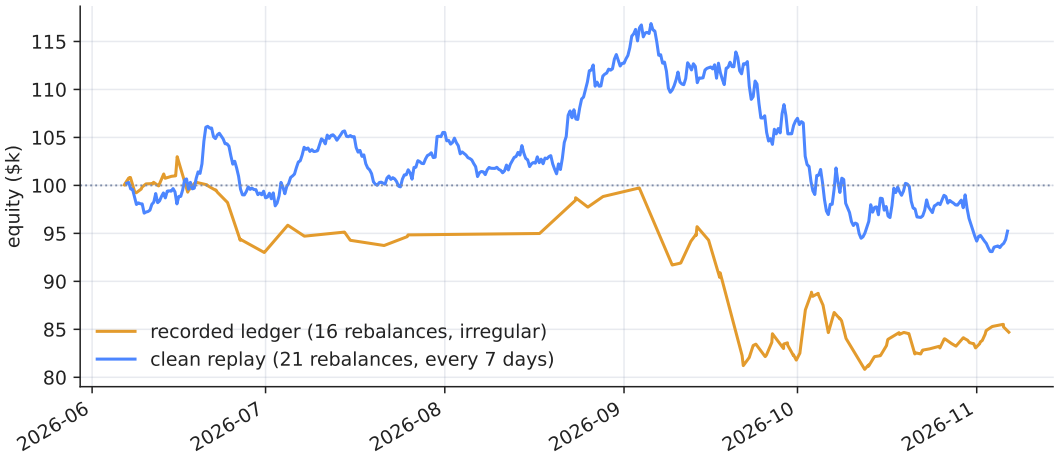


Figure 3: The same strategy over the same period, rebalanced at irregular moments (recorded) and every seven days (clean). The paths separate at rebalances, when the two books start holding different coins.

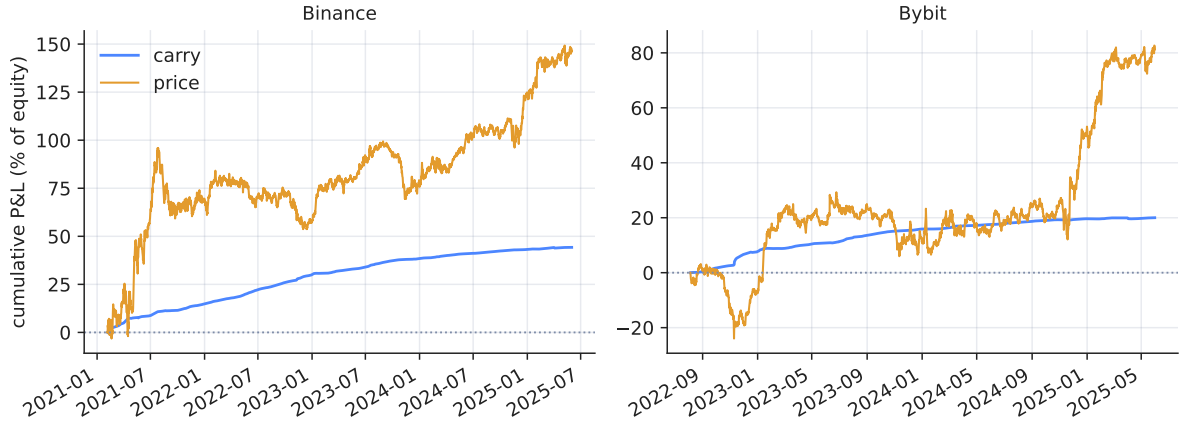


Figure 4: Baseline book, cumulative carry and price contributions in the backtests. The carry is a straight line; the price leg has the shape of a volatile directional strategy whose gains arrive in bursts.

annualised volatility below 1%.[5] The price leg, by contrast, has an annualised volatility of 30% (Binance) and 25% (Bybit). The carry leg accounts for 0.02% and 0.06% of the variance of net returns; the price leg accounts for the rest.

The backtests nonetheless look attractive because the price leg was *profitable*: 33.9% a year on Binance and 28.8% on Bybit, roughly three times the carry. Its correlation with BTC is −0.04 and 0.01, so like the live losses these gains are pure dispersion: coins with low funding outperformed coins with high funding, after removing the market.

That return is concentrated in time (Table 3). On Bybit, the best 90 days (November 2024 to February 2025) delivered 78% of the price leg's entire profit over almost three years; on Binance, April 2021 to July 2021 delivered 62%. Within those windows a handful of coins dominate. Outside them the price leg earns far less, with large swings. A backtest whose return depends this much on two or three episodes has a sampling error much larger than its Sharpe ratio suggests, and a live window that contains no such episode should be expected to do much worse than the backtest average.

[5]These carry Sharpe ratios overstate what is attainable. Funding rates are persistent, so eight-hour carry returns are strongly autocorrelated and the $\sqrt{1095}$ annualisation assumes independence. The qualitative point, that the carry is nearly riskless relative to the price leg, survives any reasonable correction.

	Binance	Bybit
Total price P&L, baseline	147%	81%
Best 90-day window	April 2021 to July 2021	November 2024 to February 2025
Price P&L in that window	90%	63%
Share of total	62%	78%
Largest contributors in that window	SAND (+16%), AXS (+15%)	DOT (+13%), COMP (+10%)

Table 3: Concentration of the price leg's backtest return. Contributions are cumulative $\sum_t w_{i,t-1} r_{i,t}$ over the window, in percent of equity.

	Binance, February 2021 to May 2025					Bybit, August 2022 to May 2025				
Variant	SR	CAGR	Carry	P. vol	MDD	SR	CAGR	Carry	P. vol	MDD
Baseline: $1/N$, BTC	1.32	42.3	10.2	30	−32.1	1.24	32.3	7.1	25	−22.8
A: inv. vol, BTC	1.46	44.5	9.6	28	−29.5	1.12	26.7	6.4	24	−19.1
B: $1/N$, BTC+ETH	1.28	38.6	10.0	29	−23.6	0.99	22.5	6.7	23	−21.6
C: inv. vol, BTC+ETH	1.48	42.3	9.3	26	−20.5	0.90	18.4	6.0	21	−20.7
D: C, re-hedged each period	1.41	39.4	9.3	26	−20.0	0.79	15.5	6.0	21	−21.2

Table 4: Weighting and hedge variants. SR: annualised Sharpe ratio. CAGR, carry (annualised), price-leg volatility and maximum drawdown in percent. All variants trade the same periods after a 35-day warm-up, with 8 bps costs on turnover.

There is an economic story under which the dispersion return is a premium rather than luck. High funding marks crowded leveraged longs, and crowded positions tend to unwind, so high-funding coins may underperform on average. The data here are too short and too concentrated to distinguish that premium from a few fortunate episodes, and I do not claim either.

5 Do better weights or a second factor help?

If the problem is residual risk, two refinements suggest themselves. *Inverse volatility weighting*: within each leg, $w_i \propto 1/\sigma_i$ with σ_i the trailing 10-day volatility, normalised to a gross of one per leg. Under equal weights the short leg, made of the most volatile coins, dominates the book's risk; inverse volatility equalises the per-name risk contributions in the spirit of risk parity [5]. *A second hedge factor*: ETH in addition to BTC, using equation (2). I also test re-hedging every period instead of only at rebalance.

Table 4 and Figure 5 give the results. Two patterns hold on both venues. Every refinement lowers the price-leg volatility (from 30% to 26% on Binance and from 25% to 21% on Bybit for variant C) and lowers the maximum drawdown. Re-hedging every period never improves the Sharpe ratio: the weekly hedge is not stale, and the extra turnover costs more than it saves.

The effect on the Sharpe ratio, however, flips sign. On Binance, variant C improves it from 1.32 to 1.48. On Bybit, the venue of the live test, it falls from 1.24 to 0.90, and each refinement makes it worse. The reason is visible in the price column of Table 4: the refinements reduce the book's dispersion exposure, and whether that raises or lowers the Sharpe ratio depends on whether the removed exposure happened to be profitable in that sample. They also cannot remove much of it. Neither BTC nor ETH spans the idiosyncratic moves of the other 21 coins, which is why the price-leg volatility falls by only 13 to 16%.

With five variants tested on two samples, a Sharpe gain of 0.16 on one venue and a loss of 0.34 on the other is not evidence of an improvement in expected return. It is evidence of a modest reduction in risk. Selecting the variant that looks best in one sample would be a textbook instance of the selection bias that deflated Sharpe ratios are designed to catch [1, 2].

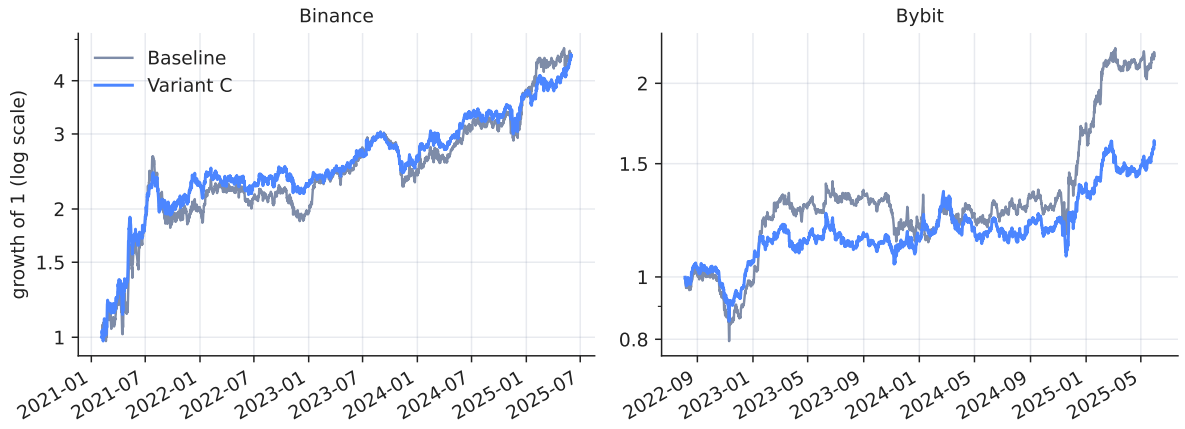


Figure 5: Growth of one unit, baseline versus variant C. Variant C is smoother on both venues; it ends higher on Binance and lower on Bybit.

6 Discussion

6.1 What the evidence supports

The book should be described as what it is: a cross-sectional dispersion strategy with a carry tailwind, not a carry strategy with residual risk. Its backtest Sharpe ratio of 1.24 to 1.32 is earned by a price leg with 25 to 30% volatility, whose return is concentrated in a few episodes and so sensitive to which coins are held that the rebalance day alone moved five months of live results by 10.5 points. On the intended schedule the live period was an ordinary bad stretch, not evidence against the backtest; five months cannot settle whether the edge is real. What the evidence does settle is that the strategy should not be run as a low-risk carry product, and that its live carry ran at about half the backtest rate.

6.2 Where the carry actually comes from

Splitting the carry by leg (baseline) is the most useful number in this study. On Binance the short leg earned +14.6% a year and the long leg −1.7%; on Bybit, +9.7% and +0.7%. The long leg contributes nothing to the carry. It is in the book only to offset the short leg's price exposure, and it does so poorly because the long and short coins are different coins.

This suggests a different construction. Shorting the high-funding perpetuals and holding the *same* coins on spot (the classic cash-and-carry trade) collects the short-leg funding while making each position delta-neutral by construction. The price leg then reduces to the spot-perpetual basis of the same coin, which is far smaller than the dispersion between different coins. This removes the dispersion bet instead of trying to hedge it with market factors. It also has costs that the cross-sectional book avoids: spot capital and custody, two venues or account types, and basis risk around funding resets. I have not tested it here; it is the next experiment.

6.3 Limitations

- *Short samples.* 4.3 and 2.8 years of backtest, and five months live. The backtest results are dominated by a few episodes, so their sampling error is larger than the Sharpe ratios suggest.
- *Shadow ledger, not fills.* The live test marks to last prices and charges no transaction costs at all; the backtests charge 8 bps on turnover. Real execution in the smaller coins would add costs and slippage, so the live figures are if anything optimistic.

- *Execution schedule.* The recorded ledger only advanced while its host machine was on (the longest gap was 23 days), so it rebalanced at irregular moments. Section 3.1 shows this moved the result by 10.5 points. The clean replay is the better estimate of the strategy as specified, but it was reconstructed after the fact rather than recorded in real time; no rule or parameter was changed to produce it. From 7 November 2026 the live job replays missed steps itself.
- *Fixed universe.* The 23 coins were chosen with hindsight, from coins that are still liquid today. That is survivorship bias in the backtest, which would flatter it.
- *Multiple testing.* The variants were chosen after seeing the live losses. Section 5 reports all of them on both venues for that reason.

6.4 Ongoing

Variant C now runs live as a second, parallel shadow ledger from 6 November 2026, while the baseline continues unchanged as a control; both now follow the clean schedule. Given Table 4, the prior is lower volatility and lower drawdown, not higher return. The quantity to watch is the price-leg volatility per step relative to the baseline, which should become measurable within weeks, long before return differences mean anything.

7 Conclusion

A funding-carry book on crypto perpetuals collects a real, stable, near-riskless cash flow and wraps it in a dispersion bet that is about thirty times more volatile. A BTC beta hedge neutralises the market well (live R^2 of 1%) and is irrelevant to the dispersion, which decided the live result: -4.8% on the intended schedule, -15.3% as recorded, a gap that comes purely from rebalancing on different days. Better weights and a second factor lower the risk modestly but do not reliably raise the return. To harvest the carry, the book has to stop holding different coins on its two sides.

Reproducibility. Code, the shadow ledger and the script that generates every number and figure in this paper (`paper/make_assets.py`) are in the `funding-carry-harvest` project of https://github.com/noaRoblesLevy/Quant-Research.

Revision note (7 November 2026). The first version reported only the recorded ledger and described its irregular schedule as a minor limitation. Replaying the period on the intended schedule showed that the schedule moved the result from -4.8% to -15.3%, that the recorded loss was a tail event mainly because of it, and that the live carry was about half the backtest rate rather than in line with it. Section 3.1 was added and the abstract, the carry and significance discussion, the limitations and the conclusion were revised. The dispersion diagnosis is unchanged.

Disclaimer. This is research, not investment advice. No real capital was traded.

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